EMCH 368 Mechatronics
Fall 2026
Practice Set 2

100 points

Name: _____ Date: _____

Complete all ten problems. Show your reasoning and use units. Unless specified otherwise, use ideal components and report calculated values to at least three significant figures. This set uses new scenarios and tasks: calibration equations, design inequalities, circuit analysis, signal filtering, and data calculations. It is a companion to Practice Set 1.

Homework connections

All references below point to `HW_fall_2026.pdf`. Page numbers are the PDF page positions, counted from 1; they are not the repeated Blackboard page numbers. Related homework questions cover the underlying concepts; these practice questions extend them.

Set 2	Topic	Related homework	PDF pp.
1	Calibration correction	HW03 Q2, Q5, Q10	13–16
2	Accuracy constraints	HW03 Q5, Q10	14–16
3	Encoder sizing	HW03 Q7, Q9	15–16
4	Light-triggered switch	HW04 Q1, Q8	17, 20–21
5	Amplifier design	HW04 Q2, Q5, Q10	17–19, 21–22
6	Sensor settling constraint	HW03 Q4; HW04 Q9	14, 21
7	Null-balance measurement	HW05 Q9	26–27
8	Filter selection	HW05 Q5, Q8	24–26
9	Multiplexed acquisition	HW06 Q4, Q8, Q10	29–32
10	Digital transmission	HW05 Q4; HW06 Q3, Q5, Q7	24, 29–30

Independent practice material based on the supplied homework. Question and solution formatting follows the supplied HW2 examples.

Problem 1. (10 pt) Build and test a calibration correction

Related homework: HW03 Q2, Q5, Q10 (PDF pp. 13–16): sensitivity and measurement error.

A pressure instrument is assumed to follow a straight-line model

$$I = mp + b,$$

where p is true pressure in kPa and I is the instrument's displayed value in kPa. A reference test gives $I = 12.0$ at $p = 0$ and $I = 392.0$ at $p = 400.0$ kPa.

(a) (3 pt) Determine m and b, and write the forward calibration equation.

(b) (3 pt) Rearrange the equation to correct any displayed value. Find the corrected pressure when the instrument displays 250.0 kPa.

(c) (4 pt) In an independent check at a true pressure of 300.0 kPa, the display reads 299.0 kPa. Calculate the uncorrected signed error and the signed error after applying your correction. Give the corrected error as a percentage of the 400.0 kPa span. Use indicated minus true for signed error.

(a) Slope, offset, and forward equation

(b) Correction equation and corrected pressure

(c) Check-point errors

Solution:

Problem 2. (10 pt) Compare two accuracy specifications

Related homework: HW03 Q5, Q10 (PDF pp. 14–16): full-span and signed measurement error.

Two pressure sensors cover 0–600 kPa. Their maximum absolute-error specifications are:

- Sensor A: $\pm 0.25\%$ of the 600 kPa span.
- Sensor B: $\pm(0.40\%$ of reading $+\ 0.200\ \mathrm{kPa})$.

For comparing the specifications, use the nominal positive pressure p as the reading in Sensor B's error formula. Add its two specified contributions directly.

(a) (3 pt) Express each maximum error magnitude E_A and $E_B(p)$ in kPa. Find the pressure where their error bounds are equal.

(b) (3 pt) Compute both error bounds at 75.0 kPa and at 500.0 kPa. Select the sensor with the smaller bound at each pressure.

(c) (4 pt) A new requirement limits maximum error to 0.500% of the nominal pressure. Find the lowest pressure where A qualifies and the lowest pressure where B qualifies. Give each qualifying interval within 0–600 kPa.

(a) Error functions and crossover

(b) Comparisons at the two pressures

(c) Minimum pressures and qualifying intervals

Solution:

Problem 3. (10 pt) Specify an encoder before buying it

Related homework: HW03 Q7, Q9 (PDF pp. 15–16).

A tracking wheel has radius 25.0 mm. Its encoder produces one counted pulse per slot per revolution. The linear distance between consecutive pulses must be no greater than 0.200 mm. Assume no slip and do not use quadrature multiplication.

(a) (4 pt) Derive the minimum slot count and round it in the direction needed to satisfy the specification.

(b) (3 pt) Available encoders have 720, 800, or 1024 slots. Choose the lowest-count model that meets the requirement and calculate its linear distance per pulse.

(c) (3 pt) Keeping your chosen encoder, what is the largest wheel radius that still meets the 0.200 mm requirement? Would a replacement wheel of radius 30.0 mm pass?

(a) Design inequality and minimum integer slot count

(b) Selected encoder and distance per pulse

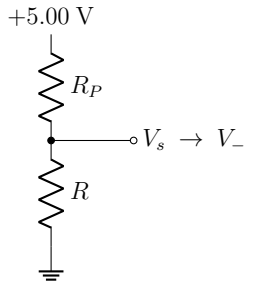
(c) Maximum radius and replacement decision

Solution:

Problem 4. (10 pt) Design a light-triggered switch

Related homework: HW04 Q1, Q8 (PDF pp. 17, 20–21).

A photoresistor R_P is connected above a fixed resistor R in a 5.00 V divider. Its midpoint is V_s. The photoresistor is $2.00\ \text{k}\Omega$ in bright light, $30.0\ \text{k}\Omega$ at the desired switching light level, and $100\ \text{k}\Omega$ in darkness. A comparator has $V_+ = 2.00$ V and $V_- = V_s$. Model its output levels as 0 and 5.00 V. Input currents are zero.



(a) (4 pt) Select R so that $V_s = 2.00$ V at the desired switching light level.

(b) (4 pt) Determine V_s and the comparator output in bright light and in darkness.

(c) (2 pt) A lamp driver turns on when the comparator output is high. Solve an inequality for the range of R_P that turns the lamp on. Exclude the exact switching threshold.

(a) Fixed resistor

(b) Bright and dark conditions

(c) Resistance range for lamp activation

Solution:

Problem 5. (10 pt) Design an amplifier and check its limits

Related homework: HW04 Q2, Q5, Q10 (PDF pp. 17–19, 21–22).

A sensor is first connected to an ideal voltage follower, then to an inverting amplifier. The desired mapping is $V_{\text{in}} = -0.300$ V to $V_o = +1.80$ V and $V_{\text{in}} = +0.900$ V to $V_o = -5.40$ V. The inverting stage has $R_i = 12.0$ kΩ. Both op-amps have output limits of ± 4.50 V. Assume input common-mode limits are not a concern.

(a) (3 pt) Determine the required linear gain and feedback resistor R_f.

(b) (3 pt) Can the full requested input range be processed without clipping? Find the largest positive sensor input that stays within the output limits, and the actual output at $+0.900$ V.

(c) (2 pt) For a separate loading test, model the sensor as a 0.450 V source in series with 6.00 kΩ. Calculate the final amplifier output with the follower present and with the follower removed (the sensor then drives R_i directly).

(d) (2 pt) At $V_{\text{in}} = +0.900$ V with the follower present, calculate the inverting-input node voltage using the actual clipped output. The non-inverting input is grounded.

(a) Gain and feedback resistor

(b) Feasibility, input limit, and clipped output

(c) Final output with and without the follower

(d) Inverting-input node voltage under clipping

Solution:

Problem 6. (10 pt) Choose a sensor for a settling deadline

Related homework: HW03 Q4; HW04 Q9 (PDF pp. 14, 21).

Two first-order temperature sensors are initially at $90.0\,^{\circ}\mathrm{C}$. Each is moved into a bath held at $30.0\,^{\circ}\mathrm{C}$. Sensor A has time constant 8.00 s; sensor B has time constant 20.0 s. Ignore dead time, noise, and steady-state error.

The acceptance requirement is: at 30.0 s after immersion, the reading must be within $2.00\,^{\circ}\mathrm{C}$ of the bath temperature.

(a) (4 pt) Calculate the remaining temperature error at 30.0 s for each sensor, and decide which sensors pass.

(b) (4 pt) Derive the largest time constant a first-order sensor could have and still pass this requirement.

(c) (2 pt) If sensor B must be used, calculate the earliest time at which its error is no greater than $2.00\,^{\circ}\mathrm{C}$.

(a) Remaining errors and pass/fail

(b) Maximum permissible time constant

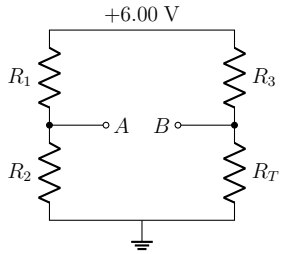
(c) Earliest acceptable measurement time for B

Solution:

Problem 7. (10 pt) Use a bridge as a null-measurement instrument

Related homework: HW05 Q9 (PDF pp. 26–27); resistor positions are defined below.

A resistance thermometer R_T is measured with a bridge. The left branch has $R_1 = 2.00\ \text{k}\Omega$ above node A and $R_2 = 3.00\ \text{k}\Omega$ below A. The right branch has adjustable R_3 above node B and R_T below B. The supply is 6.00 V, and the detector draws no current.



(a) (4 pt) The detector reads zero when $R_3 = 1.60\ \text{k}\Omega$. Determine R_T and the voltage of each midpoint relative to ground.

(b) (3 pt) For this thermometer, resistance varies linearly from $3.00\ \text{k}\Omega$ at $20.0\,^\circ\text{C}$ to $1.80\ \text{k}\Omega$ at $60.0\,^\circ\text{C}$. Find the measured temperature.

(c) (3 pt) The thermometer later reaches $60.0\,^\circ\text{C}$. What value of R_3 restores the null? With that setting and a reduced supply of 3.00 V, calculate both midpoint voltages and $V_B - V_A$.

(a) Unknown resistance and midpoint voltages

(b) Measured temperature

(c) New adjustment and midpoint voltages at 3.00 V

Solution:

Problem 8. (10 pt) Remove two different kinds of interference

Related homework: HW05 Q5, Q8 (PDF pp. 24–26).

A measurement contains useful components at 40 Hz and 200 Hz, unwanted interference at 120 Hz, and unwanted noise at 3000 Hz:

$$x(t) = 2\sin(2\pi 40t) + 0.8\sin(2\pi 120t) \\ + \sin(2\pi 200t) + 0.3\sin(2\pi 3000t) \quad [\mathrm{V}].$$

Consider the following ideal filters. Each has gain 1 in its passband and gain 0 in its stopband, with zero phase shift for this exercise.

A. Low-pass: passes frequencies below 1000 Hz.

B. High-pass: passes frequencies above 150 Hz.

C. Band-stop: rejects 100–140 Hz and passes other frequencies.

D. Band-pass: passes 100–250 Hz only.

(a) (4 pt) Choose two filters to cascade so that both useful components remain and both unwanted components are removed. Explain the purpose of each filter.

(b) (3 pt) Write the resulting output $y(t)$ under the ideal assumptions.

(c) (3 pt) Write the output signal when only filter B is used, then when only filter D is used.

(a) Selected filters and roles

(b) Output signal

(c) Outputs for B alone and D alone

Solution:

Problem 9. (10 pt) Diagnose a multiplexed data-acquisition system

Related homework: HW06 Q4, Q8, Q10 (PDF pp. 29–32).

A six-channel DAQ shares one ADC through a multiplexer. It samples channels 1 through 6 repeatedly, with one conversion per channel per scan. The ADC performs 6000 conversions/s in total. Assume equally spaced conversions and negligible switching overhead. Each channel contains desired frequencies up to 650 Hz.

(a) (3 pt) Find the sample rate of one channel and the time between successive samples of that same channel.

(b) (3 pt) A 650 Hz sinusoid appears as a lower-frequency sinusoid in the recorded data. Find its alias frequency between 0 and half the per-channel sample rate.

(c) (2 pt) To preserve the entire 0–650 Hz band, determine the strict lower bound on total conversion rate. Which available setting, 7200 or 12000 conversions/s, is sufficient under the ideal sampling criterion?

(d) (2 pt) At 12000 total conversions/s, what is the largest integer number of equally sampled channels that can each preserve a 0–650 Hz band under the same strict sampling criterion?

(a) Per-channel rate and sample interval

(b) Recorded alias frequency

(c) Total-rate requirement and selected setting

(d) Maximum channel count at 12000 conversions/s

Solution:

Problem 10. (10 pt) Decode and verify received data

Related homework: HW05 Q4; HW06 Q3, Q5, Q7 (PDF pp. 24, 29–30).

An unsigned 8-bit measurement word is transmitted with one separate even-parity bit. The data word is `01101010`. The parity bit is excluded from the numerical value.

(a) (3 pt) Decode the original data word into decimal and determine its even-parity bit.

(b) (2 pt) A receiver obtains data `01101110` with the original parity bit unchanged. Does the parity check detect an error? Show the count of 1s.

(c) (3 pt) Another transmission arrives as `01001011`, again with the original parity bit unchanged. Does this pass parity? Decode its decimal value and explain what this demonstrates.

(d) (2 pt) The ADC assigns code k to voltages in $[kq, (k+1)q)$, with $q = 0.0200$ V. It reports the midpoint $(k+0.5)q$ as its voltage estimate. For the original word, calculate the input-voltage interval, midpoint estimate, and maximum quantization-error magnitude.

(a) Original decimal value and parity bit

(b) First received word

(c) Second received word

(d) Voltage interval, estimate, and error magnitude

Solution: